SANKALP KAILASH WANJARI

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SUMMARY

AI Engineer with 3 years of software engineering experience, focused on AI agent safety, reliability, and empirical evaluation. Built agentic systems for prompt-injection defense, task-scoped authorization, and evidence-grounded research. Experienced in adversarial testing, controlled evaluations, LLM fine-tuning (LoRA/DPO), and developing reproducible ML systems using Python.

SKILLS

AI Safety & Evals	Prompt Injection, AI Control, Adversarial Testing, Agent Monitoring, Task-Scoped Authorization, LLM Evaluations, Benchmarking, Safety-Utility Trade-offs
ML & Post-Training	PyTorch, Hugging Face Transformers, Supervised Fine-Tuning, LoRA/QLoRA, Direct Preference Optimization (DPO), Model Evaluation, Error Analysis
Agents & Retrieval	LangGraph, Multi-Agent Systems, Tool Calling, RAG, Hybrid Search, BM25, Qdrant, RRF, Cross-Encoder Reranking, NLI Verification
Programming & Data	Python, FastAPI, REST APIs, SQL, C#, ASP.NET Core, PostgreSQL, Redis, React, TypeScript
Research & Tools	Experimental Design, Baseline Comparisons, Reproducible Evaluations, Git, Docker, GitHub Actions, CI/CD, Model Tracing

WORK EXPERIENCE

Software Engineer (Implementation-Conversion Analyst), FIS Global **Jun 2022 – Jul 2025**
Bangalore, India

- **Built** an end-to-end LLM SQL optimizer using **ASP.NET Core/C#, Angular, SQL Server, and an enterprise AI API**, generating and evaluating **3 candidate rewrites per request** against execution-plan cost, CPU, I/O, memory, and row counts, with structured JSON validation and penalty-based fallbacks to select the lowest-cost valid query, reducing estimated execution time by ∼**35%**.
- **Improved** secure document-processing reliability by building a **C#/.NET REST service** for AES-based encryption/decryption, integrating **Azure Blob storage** and streamed file responses, and implementing encryption-state detection and custom HTTP status metadata to prevent double processing and support large-file workflows.
- **Developed** internal banking web-application workflows using **Angular, .NET, SQL Server, and stored procedures**, supporting project tracking, request management, filtering, pagination, and migration operations across workloads containing **millions to billions of financial records**.
- **Reduced** manual document-data entry by **90%** while maintaining a **<5% correction rate** by engineering a configuration-driven extraction framework supporting **50+ financial-document layouts** without layout-specific code changes.
- **Resolved** PDF-processing failures across **1,000+ regulated financial documents** by tracing incompatible encryption metadata in raw PDF structures and converting the diagnosis into a reusable **Python automated detection utility**.

PROJECTS

IntentSeal — Prompt-Injection Defense & AI Agent Control *(Python, LLM Security, Adversarial Evals)* GitHub | PyPI

- Architected a **prompt-injection defense and AI control gateway** for autonomous agents, combining multi-tier content inspection with **deterministic, task-scoped authorization** to prevent untrusted documents, emails, and retrieved context from triggering unauthorized tool actions.
- Evaluated adversarial robustness across multiple indirect attack channels, measuring **99.2% detection recall on 359 delivered third-party attacks** and a **5.1% false-positive rate** on 39 held-out hard-benign cases to quantify safety-utility trade-offs.
- Conducted **paired protected-versus-unprotected agent evaluations**, reducing end-to-end compromises from **7/27 to 0/27**; performed an **ablation study** isolating the deterministic action-control layer with all input detectors disabled, recording **0/28 successful attacks**.
- Published the open-source **IntentSeal Python SDK and core engine on PyPI**, enabling embedded and gateway-based policy enforcement across MCP tool boundaries and existing agent frameworks.

Veriscope — Verifiable AI Research Platform *(Python, FastAPI, LangGraph, Qdrant, PostgreSQL, Redis, Docker)* GitHub

- Improved retrieval recall from **37% to 87%** using BM25, dense vector search, reciprocal-rank fusion, and cross-encoder reranking across a 28-paper evaluation corpus.
- Cut fabricated citations to **0** while maintaining **98%+ claim support** and **95%+ refusal correctness** through exact-quote matching and NLI verification against a frozen golden dataset.
- Built a 9-node LangGraph workflow with provider failover, circuit breaking, and adversarial defenses detecting **6/6 tested attack patterns with 0 false positives**, validated across a **479-test backend suite**.
- Configured a 7-job GitHub Actions pipeline covering backend validation, frontend builds, Playwright tests, security checks, container scanning, image release, and conditional deployment.

StructAgent — Fine-Tuned Structured Extraction *(Python, PyTorch, Hugging Face, QLoRA, DPO, FastAPI, Docker)* GitHub

- Engineered an end-to-end post-training pipeline fine-tuning Qwen2.5-1.5B via **LoRA SFT and DPO alignment** on a single 6 GB GPU, improving schema compliance from 25.0% to **99.0%** and exact match from 11.0% to **99.0%**.
- Reduced model footprint by **75%** using QLoRA 4-bit NormalFloat quantization, fitting memory within 6 GB VRAM while maintaining **100% tool selection accuracy** and a **0.0% hallucination rate** across 100 held-out evaluation samples.
- Built an asynchronous FastAPI inference microservice with client-side text extraction, returning standardized JSON envelopes with latency, schema validation, and real-time schema repair metadata.
- Developed an in-memory AI observability pipeline exposing cumulative token metrics, latency distributions, and schema compliance tracking via FastAPI and Prometheus endpoints.

ACHIEVEMENTS

- **7th globally - HackerRank Orchestrate**, 24-hour AI Agent hackathon, 35,712 registrants. **View Leaderboard**

EDUCATION

International Institute of Information Technology, Pune **2018 – 2022**
Bachelor of Computer Engineering with Honors in AI/ML CGPA: 8.54

EXECUTIVE EDUCATION

BITS School of Management (BITSoM) **2025 – 2026**
Executive Certificate in **Product Management with Generative & Agentic AI**